



VECTOR SPACES

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CLASS 7 : CONTENT



➤ Null Space

NULL SPACE

Definition :

Let A be a matrix of order $m \times n$.

The **null space of A** is the set of all solutions of the homogeneous system of equations

$Ax = 0$ denoted by $N(A)$.

Thus,

$$N(A) = \{ x \in R^n / Ax = 0 \}$$

Note : $N(A)$ is a subspace of R^n .

NULL SPACE

- Null Space of a matrix A is denoted by $N(A)$
- Null Space of A is spanned by Special solutions to $Ax=0$ which is same as solving for $Rx=0$ where R is the row reduced echelon form of A
- Null space of A is a subspace of vector space $\mathbb{R}^n$
- Dimension of Null space is 'n-r'
- For a system of linear equations to be nonsingular and matrix A to be invertible $N(A) = \{0\}$
- Special solutions are the basis of $N(A)$.

NULL SPACE

Example :

Let $A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 \\ 0 & \textcircled{4} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \textcircled{x} \\ \textcircled{y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(x, y both are pivot variables)
No Free variables.

Then gives $x = y = 0$ as the only solution.

The null space of this matrix thus contains only the zero vector $(0, 0)$.

Null space of this matrix is 'origin' in $\mathbb{R}^2$.

NULL SPACE

$$\begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$(x = -y = z)$ hence

Gives infinitely many solutions $(c, -c, c)$ all of which lie on a line that obviously passes through the origin.

The matrices $A = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 3 & 5 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 3 \end{bmatrix}$

have the same column space but different null space !!



THANK YOU

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